

JavaScript Assignment – Functional Programming & Design Patterns

Objective: Practice functional programming concepts and common JavaScript design patterns by implementing real-world scenarios.

Q1 Pure Functions & Immutability

Create `addItem`, `removeItem` and `updateQuantity` that return new cart objects without mutating the original.

Q2 Function Composition

Implement `pipe()` and process a product name: trim, lowercase, capitalize, add status='Available'.

Q3 Module Pattern

Create a Bank Account module with private balance and `deposit`, `withdraw`, `getBalance` methods.

Q4 Singleton Pattern

Create a Logger singleton. Verify multiple `getInstance()` calls return the same object.

Q5 Observer / Pub-Sub

Implement `EventBus` with `subscribe/publish`. Notify Inventory, Email and Analytics when an order is placed.

Q6 Factory Pattern

Create a Payment Factory supporting Credit Card, UPI and Cash with `pay(amount)`.

Q7 Decorator Pattern

Implement `withExecutionTime()` to measure execution time of any function.

Q8 Mini Project

Build a Food Delivery Order System using Pure Functions, Composition, Singleton, Factory, Observer and Decorator patterns.

Bonus

+10: Build a Library Management System using all the above concepts.

Submission: Clean JavaScript, meaningful names, comments, sample input/output, separate JS files or one project. Total: 100 +10 bonus.